

APS Spectral Flow for Selective Propagation in Geometric Deep Learning

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Abstract

We develop a rigorous core for a selective spectral-flow approach to structured operator models motivated by Jacobian-chain propagation. Starting from a scalar Atiyah–Patodi–Singer index lemma on the interval, we prove four explicit families of results: an S^1 -equivariant compact model, an $SO(3)$ -equivariant spherical model on S^2 , a continuous-spectrum semifinite model, and a radial hyperbolic model on real hyperbolic space. In all these settings, nuisance channels satisfying nonresonance or uniform-gap conditions contribute no protected propagation, whereas task channels contribute through explicit sector- or mode-resolved crossing counts. The present paper does not yet establish a full proper-action Jacobian-chain index theory. Rather, it provides a rigorous core and a family of computable structured examples for a broader program relating selective spectral flow, representation-resolved propagation, continuous-mode spectral-flow densities, and no-crossing theorems in geometric deep learning.

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1 Introduction

1.1 The deep-learning problem

One of the persistent conceptual questions in deep learning is how to distinguish information that propagates stably through a model from information that is systematically suppressed, collapsed, or eliminated. In structured settings — for instance, in equivariant models, on homogeneous spaces, and on non-Euclidean state spaces — this question is naturally geometric. The geometric deep learning program explicitly treats groups, homogeneous spaces, and harmonic analysis as organizing principles for both analysis and design [BBCV21].

From the viewpoint of representation learning, the same issue may be phrased as the need to suppress nuisance variation while preserving task-relevant information. This viewpoint appears explicitly in the invariance/disentanglement literature [AS18] and operationally in modern equivariant architectures, where symmetry constraints are introduced precisely because nuisance transformations should not corrupt the task signal [FWFW20]. The problem addressed in this paper is whether, in structured settings of this kind, this task/nuisance distinction can be turned into a rigorous propagation theory.

1.2 Why the problem naturally enters our mathematical framework

In harmonic-analysis-friendly models, depth can often be encoded by a one-parameter family of selfadjoint operators

$$u \longmapsto A_u, \quad u \in [0, 1],$$

and the state space can often be decomposed either into irreducible representation sectors or into continuous spectral modes. This is natural in several ways. First, continuous-depth viewpoints are already standard in deep learning: residual architectures admit continuous-depth idealizations and ODE-type formulations [CRBD18]. Second, equivariant models on the circle, sphere, or in Euclidean 3D naturally decompose into Fourier, spherical, or representation-theoretic sectors

[CGKW18, EABMD18, FWW20]. Third, hyperbolic and more general noncompact symmetric-space models naturally carry a Helgason–Fourier / spherical-transform decomposition, and these have already appeared in the deep-learning literature [GBH18, SII22].

Once the model is decomposed into sectors or modes, the depth-evolution problem becomes a spectral-crossing problem. A task channel is allowed to cross zero and hence contribute protected propagation. A nuisance channel should satisfy a structural condition — such as nonresonance in compact sector models, or a uniform positive gap in continuous-spectrum models — that prevents any spectral crossing. Atiyah–Patodi–Singer type index formulas and spectral flow are therefore natural tools for counting net crossings.

1.3 Relation to the index-theoretic and geometric-DL literature

The index-theoretic background comes from two sources. On the one hand, Lichnerowicz-type vanishing results show that positivity can force indices to vanish. For the proper cocompact action setting relevant to noncompact manifolds, a published Lichnerowicz-type vanishing theorem is due to Weiping Zhang [Zha17]. On the other hand, proper-action equivariant index theory in the sense of Hochs–Song identifies the natural target of a full general theory as a class in

$$KK(C_0(G/K) \rtimes G, \mathbb{C}),$$

with Morita identification to $\widehat{R}(K)$ and an induction picture via global K -slices [HS17, HS18]. On the analytical side, our continuous-spectrum arguments rely on the semifinite spectral-flow framework of Azzali–Wahl [AW11], while the eventual $K_0(J)$ -valued refinement is suggested by Kaad–Nest–Rennie [KNR12].

On the deep-learning side, our work is closest in spirit to the geometric and equivariant-learning literature. Spherical CNNs supply a canonical compact nonabelian example in which harmonic-analysis sector decompositions are central [CGKW18, EABMD18]. SE(3)-Transformers show how symmetry constraints and attention combine in 3D settings where nuisance transformations are central [FWW20]. Hyperbolic neural networks provide standard evidence that non-Euclidean geometry matters operationally for representation learning [GBH18]. Most directly relevant for our noncompact setting, Sonoda–Ishikawa–Ikeda use Helgason–Fourier analysis to build fully connected networks on noncompact symmetric spaces [SII22]. At a more conceptual level, the geometric deep learning blueprint emphasizes that such structures should organize both analysis and design [BBCV21], while the topological deep learning position paper explicitly lists theoretical foundations as a central open problem [PBB⁺24].

Relative to these works, our contribution is not to introduce another structured neural architecture. Rather, our contribution is to isolate propagation invariants and no-contribution theorems for structured operator models motivated by Jacobian-chain propagation: representation-resolved propagation counts in compact settings, mode-resolved spectral-flow densities in continuous-spectrum settings, and strict nuisance no-crossing results under verifiable structural conditions.

1.4 What is new in this paper

The paper proves a rigorous core consisting of four kinds of results. First, starting from a scalar APS lemma on the interval, we build a compact S^1 -equivariant model in which the equivariant index is an explicit sum of Fourier-sector crossing counts. Second, we upgrade this to a genuinely nonabelian compact homogeneous space by proving an $SO(3)$ -equivariant spherical theorem on S^2 . Third, we treat a continuous-spectrum semifinite model and prove that the total Breuer spectral flow is the integral of an explicit task spectral-flow density. Fourth, we place the noncompact theory into a genuinely geometric setting by proving a radial hyperbolic theorem in which the symbol is dictated by the spectrum of $(-\Delta_{\mathbb{H}^d})^{1/2}$ rather than by an abstract toy multiplier. Across all four settings, nuisance channels satisfying nonresonance or gap hypotheses

contribute nothing. This selective vanishing mechanism is the central structural point of the paper.

1.5 Main theorem architecture of the paper

To avoid any ambiguity, all principal mathematical statements are gathered in Section 2 as Theorems 2.1–2.5. Section 3 records the external input and conventions. Section 4 proves the scalar APS lemma. Section 5 proves the compact S^1 model. Section 6 proves the continuous-spectrum semifinite model. Section 7 proves the nonabelian compact spherical model on S^2 . Section 8 proves a weighted Plancherel theorem and then applies it to a radial hyperbolic model.

1.6 Scope of the present paper

This paper does *not* establish a full proper-action Jacobian-chain index theory, does *not* claim a theorem for arbitrary neural architectures, and does *not* derive a practical optimizer or training recipe for modern large-scale models. What it does provide is a rigorous core for a broader program: a family of fully proved compact and noncompact structured theorems showing how task-relevant channels can be counted by spectral crossings while nuisance channels are forced into a no-crossing regime. That is the precise mathematical contribution of the present paper.

2 Main Results

This section gathers the principal statements proved in the paper. The purpose of collecting them here is twofold. First, it makes the mathematical architecture of the paper explicit from the outset. Second, it clarifies for the deep-learning reader exactly what propagation problem is being solved in each setting.

2.1 The scalar propagation-counting theorem

Theorem 2.1. *Let*

$$c \in C^1([0, 1], \mathbb{R}), \quad c(0) \neq 0, \quad c(1) \neq 0.$$

Let

$$b_c = \frac{d}{du} + c(u)$$

be the scalar APS operator on $L^2([0, 1])$ with the APS boundary conditions determined by the signs of $c(0)$ and $c(1)$. Then

$$\text{ind}(b_c) = \frac{1}{2}(\text{sgn } c(1) - \text{sgn } c(0)).$$

If c has only simple zeros in $(0, 1)$, then

$$\text{ind}(b_c) = \sum_{u_0: c(u_0)=0} \text{sgn } c'(u_0).$$

Deep-learning interpretation. This theorem is the one-dimensional atomic propagation-counting rule. For a single scalar channel evolving with depth parameter u , the net protected propagation is exactly the net number of signed zero-crossings. Every later theorem is obtained by decomposing a structured model into channels or modes and summing copies of this scalar counting law.

2.2 A compact representation-resolved theorem on the circle

Theorem 2.2. *Consider the compact homogeneous model on S^1 defined in Section 5. Then the associated S^1 -equivariant APS operator B is Fredholm and satisfies*

$$\text{Ind}_{S^1}(B) = \sum_{n \in \mathbb{Z}} \left(\sum_{j=1}^{r_T} \sigma_{n,j}^T + \sum_{k=1}^{r_N} \sigma_{n,k}^N \right) [\chi_n],$$

where

$$\sigma_{n,j}^T = \frac{1}{2} \left(\text{sgn}(n + \alpha_j(1)) - \text{sgn}(n + \alpha_j(0)) \right), \quad \sigma_{n,k}^N = \frac{1}{2} \left(\text{sgn}(n + \beta_k(1)) - \text{sgn}(n + \beta_k(0)) \right).$$

If the nuisance channels satisfy

$$-\beta_k([0, 1]) \cap \mathbb{Z} = \emptyset \quad (1 \leq k \leq r_N),$$

then

$$\text{Ind}_{S^1}(B) = \sum_{n \in \mathbb{Z}} \left(\sum_{j=1}^{r_T} \sigma_{n,j}^T \right) [\chi_n].$$

Deep-learning interpretation. The propagation problem is not global but symmetry-resolved. The integer n labels the irreducible S^1 -sector. Task and nuisance channels are counted inside each sector separately. The nonresonance hypothesis is a strict nuisance no-contribution theorem for protected propagation.

2.3 A nonabelian compact theorem on the sphere

Theorem 2.3. *Consider the $SO(3)$ -equivariant spherical model on S^2 defined in Section 7. Then the associated APS operator B_{S^2} is $SO(3)$ -Fredholm and satisfies*

$$\text{Ind}_{SO(3)}(B_{S^2}) = \sum_{\ell=0}^{\infty} \left(\sum_{j=1}^{r_T} \sigma_{\ell,j}^T + \sum_{k=1}^{r_N} \sigma_{\ell,k}^N \right) [V_\ell],$$

where V_ℓ is the irreducible $SO(3)$ -representation of degree ℓ and

$$\sigma_{\ell,j}^T = \frac{1}{2} \left(\text{sgn}(p_{j,\ell} + \alpha_j(1)) - \text{sgn}(p_{j,\ell} + \alpha_j(0)) \right),$$

$$\sigma_{\ell,k}^N = \frac{1}{2} \left(\text{sgn}(q_{k,\ell} + \beta_k(1)) - \text{sgn}(q_{k,\ell} + \beta_k(0)) \right).$$

If

$$-q_{k,\ell} \notin \beta_k([0, 1]) \quad (\forall k, \forall \ell),$$

then all nuisance contributions vanish. In particular, the theorem applies to geometric spectral-filter families built from the spherical Laplacian $-\Delta_{S^2}$.

Deep-learning interpretation. This theorem is the nonabelian compact analogue of Theorem 2.2. Instead of Fourier modes indexed by $n \in \mathbb{Z}$, one now has spherical harmonic channels indexed by irreducible representations V_ℓ of $SO(3)$. Protected propagation can be counted representation-by-representation, and nuisance channels can be excluded representation-by-representation as well.

2.4 A continuous-spectrum selective spectral-flow theorem

Theorem 2.4. *Consider the continuous-spectrum semifinite model of Section 6. Then the associated selfadjoint path $(A_u)_{u \in [0,1]}$ has well-defined Breuer spectral flow and*

$$\text{sf}_{\text{Tr}_{\text{Pl}}}\{A_u\}_{u \in [0,1]} = \int_{\mathbb{R}} \nu_{\text{T}}(\lambda) d\lambda = \sum_{j=1}^{r_{\text{T}}} (\alpha_j(1) - \alpha_j(0)),$$

where

$$\nu_{\text{T}}(\lambda) = \sum_{j=1}^{r_{\text{T}}} \frac{1}{2} \left(\text{sgn}(\lambda + \alpha_j(1)) - \text{sgn}(\lambda + \alpha_j(0)) \right).$$

All nuisance channels contribute trivially. More generally, in the weighted-measure model of Section 8,

$$\text{sf}_{\text{Tr}_{\mu}}\{A_u^{\mu}\}_{u \in [0,1]} = \int_{\mathbb{R}} \nu_{\text{T}}^{\mu}(\lambda) d\mu(\lambda) = \sum_{j=1}^{r_{\text{T}}} \mu(I_j^+) - \sum_{j=1}^{r_{\text{T}}} \mu(I_j^-).$$

Deep-learning interpretation. This theorem replaces discrete symmetry sectors by continuous spectral modes. The quantity $\lambda \mapsto \nu_{\text{T}}(\lambda)$ is a mode-resolved propagation density. It says precisely where task-relevant propagation occurs in continuous spectrum. The nuisance channels are uniformly gapped and therefore carry no protected propagation at any mode.

2.5 A genuinely geometric noncompact theorem on real hyperbolic space

Theorem 2.5. *Consider the radial hyperbolic model on*

$$\mathbb{H}^d = SO_0(d, 1)/SO(d)$$

defined in Section 8, where the symbol is dictated by the spectrum of

$$(-\Delta_{\mathbb{H}^d})^{1/2}$$

through

$$s_{\rho}(\lambda) = \sqrt{\lambda^2 + \rho^2}, \quad \rho = \frac{d-1}{2}.$$

Then the associated semifinite spectral flow is

$$\text{sf}_{\text{Tr}_{\rho}}\{A_u^{\rho}\}_{u \in [0,1]} = \int_0^{\infty} \nu_{\text{T}}^{\rho}(\lambda) d\mu_{\rho}(\lambda) = \sum_{j=1}^{r_{\text{T}}} \mu_{\rho}(J_j^+) - \sum_{j=1}^{r_{\text{T}}} \mu_{\rho}(J_j^-),$$

where

$$\nu_{\text{T}}^{\rho}(\lambda) = \sum_{j=1}^{r_{\text{T}}} \frac{1}{2} \left(\text{sgn}(\tau_j(1) - s_{\rho}(\lambda)) - \text{sgn}(\tau_j(0) - s_{\rho}(\lambda)) \right).$$

Again, nuisance channels contribute trivially.

Deep-learning interpretation. This is the genuinely geometric noncompact version of the selective spectral-flow principle. The crossing variable is no longer an abstract toy spectral parameter but the true spherical spectral variable of real hyperbolic space. Thus task-relevant propagation is mode-resolved by the hyperbolic Plancherel spectrum, while nuisance channels are removed by a strict no-crossing condition.

2.6 What these theorems jointly show

Taken together, Theorems 2.1–2.5 establish a single structural principle:

task channels contribute through signed spectral crossings, while nuisance channels satisfying nonresonance or gap conditions contribute no protected propagation.

The compact and noncompact settings differ only in how the channels are organized: by irreducible representations in the compact case, and by continuous spectral modes in the noncompact case. The remainder of the paper is devoted to proving these statements.

3 External input and standing conventions

Only a very small amount of external machinery is used. We isolate it here.

3.1 External input

- (A) **Zhang’s proper-action Lichnerowicz vanishing result.** For proper cocompact actions on noncompact spin manifolds, a published Lichnerowicz-type vanishing theorem is due to Weiping Zhang. It provides a proper-action vanishing statement for the relevant Mathai–Zhang index and serves as our vanishing prototype [Zha17].
- (B) **Hochs–Song proper-action index.** For proper actions, one has an equivariant index class

$$\text{index}_G(D) \in KK(C_0(G/K) \rtimes G, \mathbb{C}),$$

together with a Morita identification with $\widehat{R}(K)$ and an induction picture via K -slices [HS17, HS18].

- (C) **Azzali–Wahl semifinite spectral flow = APS index.** Let N be a semifinite von Neumann algebra with a faithful normal semifinite trace. For a common-domain selfadjoint path $(D_u)_{u \in [0,1]}$ such that $(D_u \pm i)^{-1} \in K(N)$ and the path is constant near the endpoints, the semifinite spectral flow is well-defined and equals the Breuer index of the corresponding APS operator $(\partial_u + D_u)_{\text{APS}}$ [AW11].
- (D) **$K_0(J)$ -valued spectral flow.** For a norm-closed ideal J in a von Neumann algebra, spectral flow may be refined to a class in $K_0(J)$; in the semifinite setting, the numerical spectral flow is obtained by applying the trace to that class [KNR12].
- (E) **Classical spherical transform on real hyperbolic space.** For real hyperbolic space

$$\mathbb{H}^d = SO_0(d, 1)/SO(d), \quad \rho = \frac{d-1}{2},$$

there is a unitary spherical transform on the radial subspace

$$\mathcal{F}_{\text{sph}} : L^2_{\text{rad}}(\mathbb{H}^d) \longrightarrow L^2([0, \infty), d\mu_\rho),$$

where

$$d\mu_\rho(\lambda) = |c(\lambda)|^{-2} d\lambda,$$

which diagonalises $(-\Delta_{\mathbb{H}^d})^{1/2}$ as multiplication by

$$\sqrt{\lambda^2 + \rho^2}$$

[Cam90, Hel00].

3.2 Standing conventions

Throughout the paper, the evolution parameter is denoted by $u \in [0, 1]$. Compact symmetry groups are denoted by K . A compact homogeneous slice is denoted by Σ . Task and nuisance sectors are indexed by T and N. For a closed operator T , the notation $H(T)$ refers to its domain equipped with the graph norm.

Except for External inputs (C) and (E), all results in Sections 4–8 are proved in the present paper.

4 A scalar APS lemma on the interval

This section provides the basic one-dimensional building block used in all later calculations.

Definition 4.1. Let $c \in C^1([0, 1], \mathbb{R})$ satisfy $c(0) \neq 0$ and $c(1) \neq 0$. Define

$$\varepsilon_0 := \mathbf{1}_{\{c(0) > 0\}}, \quad \varepsilon_1 := \mathbf{1}_{\{c(1) < 0\}}.$$

The scalar APS operator associated with c is

$$b_c := \frac{d}{du} + c(u)$$

on $L^2([0, 1])$ with domain

$$\text{Dom}(b_c) := \{f \in H^1([0, 1]) : \varepsilon_0 f(0) = 0, \varepsilon_1 f(1) = 0\}.$$

Lemma 4.2. *Let*

$$d_{\varepsilon_0, \varepsilon_1} : \text{Dom}(b_c) \rightarrow L^2([0, 1]), \quad d_{\varepsilon_0, \varepsilon_1} f = f'.$$

Then $d_{\varepsilon_0, \varepsilon_1}$ is Fredholm, and the following hold.

1. *If $(\varepsilon_0, \varepsilon_1) = (1, 0)$, then $d_{\varepsilon_0, \varepsilon_1}$ is bijective, with inverse*

$$(d_{\varepsilon_0, \varepsilon_1}^{-1} g)(u) = \int_0^u g(s) ds.$$

2. *If $(\varepsilon_0, \varepsilon_1) = (0, 1)$, then $d_{\varepsilon_0, \varepsilon_1}$ is bijective, with inverse*

$$(d_{\varepsilon_0, \varepsilon_1}^{-1} g)(u) = - \int_u^1 g(s) ds.$$

3. *If $(\varepsilon_0, \varepsilon_1) = (0, 0)$, then*

$$\ker d_{\varepsilon_0, \varepsilon_1} = \mathbb{C}, \quad \text{Ran } d_{\varepsilon_0, \varepsilon_1} = L^2([0, 1]),$$

hence $\text{ind}(d_{\varepsilon_0, \varepsilon_1}) = 1$.

4. *If $(\varepsilon_0, \varepsilon_1) = (1, 1)$, then*

$$\ker d_{\varepsilon_0, \varepsilon_1} = 0, \quad \text{Ran } d_{\varepsilon_0, \varepsilon_1} = \left\{ g \in L^2([0, 1]) : \int_0^1 g(s) ds = 0 \right\},$$

hence $\text{ind}(d_{\varepsilon_0, \varepsilon_1}) = -1$.

In particular,

$$\text{ind}(d_{\varepsilon_0, \varepsilon_1}) = \frac{1}{2} (\text{sgn } c(1) - \text{sgn } c(0)).$$

Proof. If $(\varepsilon_0, \varepsilon_1) = (1, 0)$, then the domain consists of functions in $H^1([0, 1])$ vanishing at 0. Every $g \in L^2([0, 1])$ has a primitive

$$f(u) = \int_0^u g(s) ds$$

with $f \in H^1([0, 1])$, $f(0) = 0$, and $f' = g$. Hence the operator is surjective. If $f' = 0$ and $f(0) = 0$, then $f \equiv 0$, hence injective.

The case $(\varepsilon_0, \varepsilon_1) = (0, 1)$ is identical with the inverse

$$f(u) = - \int_u^1 g(s) ds.$$

If $(\varepsilon_0, \varepsilon_1) = (0, 0)$, the domain is all of $H^1([0, 1])$. The kernel consists of constant functions, so it is one-dimensional. Every $g \in L^2([0, 1])$ has a primitive in $H^1([0, 1])$, hence the operator is surjective.

If $(\varepsilon_0, \varepsilon_1) = (1, 1)$, the domain is $H_0^1([0, 1])$. Its kernel is trivial, since a constant function vanishing at both endpoints is zero. If $g = f'$ with $f \in H_0^1([0, 1])$, then

$$\int_0^1 g(s) ds = f(1) - f(0) = 0.$$

Conversely, every zero-integral $g \in L^2([0, 1])$ has a primitive in $H_0^1([0, 1])$. Hence the range is exactly the codimension-one zero-integral hyperplane. The final formula follows by inspecting the sign cases. $\square$

Lemma 4.3. *The Hilbert-space adjoint of b_c is*

$$b_c^* = -\frac{d}{du} + c(u),$$

with domain

$$\text{Dom}(b_c^*) = \{g \in H^1([0, 1]) : (1 - \varepsilon_0)g(0) = 0, (1 - \varepsilon_1)g(1) = 0\}.$$

Proof. For $f \in \text{Dom}(b_c)$ and $g \in H^1([0, 1])$, integration by parts gives

$$\langle b_c f, g \rangle - \langle f, (-g' + cg) \rangle = f(1)\overline{g(1)} - f(0)\overline{g(0)}.$$

To annihilate this boundary term for all $f \in \text{Dom}(b_c)$, one must impose the stated complementary endpoint conditions. $\square$

Theorem 4.4. *The scalar APS operator b_c is Fredholm and satisfies*

$$\text{ind}(b_c) = \frac{1}{2}(\text{sgn } c(1) - \text{sgn } c(0)).$$

Proof. Let

$$h(u) := \int_0^u c(s) ds, \quad (M_h f)(u) := e^{h(u)} f(u).$$

Since h is continuous, M_h and M_h^{-1} are bounded on $L^2([0, 1])$, preserve $H^1([0, 1])$, and preserve the APS boundary conditions because $e^{h(0)}$ and $e^{h(1)}$ are nonzero. A direct computation gives

$$M_h b_c M_h^{-1} = d_{\varepsilon_0, \varepsilon_1}.$$

Hence b_c is similar to $d_{\varepsilon_0, \varepsilon_1}$ and therefore Fredholm. The index formula follows from Lemma 4.2. $\square$

Corollary 4.5. *If c has only finitely many simple zeros $u_1, \dots, u_N \in (0, 1)$, then*

$$\text{ind}(b_c) = \sum_{i=1}^N \text{sgn } c'(u_i).$$

Proof. Each simple zero contributes $+1$ if the sign changes from negative to positive and -1 if the sign changes from positive to negative. The total signed number of such sign changes is therefore the sum on the right-hand side. This total signed number equals

$$\frac{1}{2}(\text{sgn } c(1) - \text{sgn } c(0)),$$

so the statement follows from Theorem 4.4. $\square$

Corollary 4.6. *Let $c \in C^1([0, 1], \mathbb{R})$ with $c(0) \neq 0$ and $c(1) \neq 0$. Then*

$$\dim \ker b_c = \begin{cases} 1, & c(0) < 0 < c(1), \\ 0, & \text{otherwise,} \end{cases} \quad \dim \ker b_c^* = \begin{cases} 1, & c(0) > 0 > c(1), \\ 0, & \text{otherwise.} \end{cases}$$

Proof. This is exactly the explicit kernel and adjoint-kernel computation contained in the proof of Theorem 4.4, together with Lemma 4.3. $\square$

5 A compact homogeneous S^1 -model

This section provides a fully explicit compact model in which the equivariant index decomposes into irreducible S^1 -sectors and the task/nuisance splitting can be read off sector by sector.

5.1 Definition of the model

Let

$$K = S^1, \quad \Sigma = S^1,$$

where K acts on Σ by rotation. Let

$$r_T, r_N \in \mathbb{N} \cup \{0\}, \quad m := r_T + r_N.$$

Set

$$\mathcal{H} := L^2(S^1, \mathbb{C}^m), \quad \mathcal{D} := H^1(S^1, \mathbb{C}^m).$$

Let

$$\alpha_j, \beta_k \in C^1([0, 1], \mathbb{R}),$$

for $1 \leq j \leq r_T$ and $1 \leq k \leq r_N$. Assume:

1. each α_j and β_k is constant near $u = 0$ and near $u = 1$;
- 2.

$$\alpha_j(0), \alpha_j(1), \beta_k(0), \beta_k(1) \notin \mathbb{Z}$$

for all j, k .

Define

$$A_u = \bigoplus_{j=1}^{r_T} \left(-i \frac{d}{d\theta} + \alpha_j(u) \right) \oplus \bigoplus_{k=1}^{r_N} \left(-i \frac{d}{d\theta} + \beta_k(u) \right).$$

Lemma 5.1. *For each $u \in [0, 1]$, the operator A_u is selfadjoint on the common domain*

$$\mathcal{D} = H^1(S^1, \mathbb{C}^m).$$

Moreover, for all $u, v \in [0, 1]$, the difference $A_u - A_v$ is bounded on $L^2(S^1, \mathbb{C}^m)$.

Proof. The operator $-i \frac{d}{d\theta}$ is selfadjoint on $H^1(S^1)$. Each scalar summand is a bounded selfadjoint perturbation, hence selfadjoint on $H^1(S^1)$. Finite orthogonal direct sums preserve selfadjointness. The difference $A_u - A_v$ is a diagonal multiplication by the bounded scalars $\alpha_j(u) - \alpha_j(v)$ and $\beta_k(u) - \beta_k(v)$. $\square$

Definition 5.2. Let T be an S^1 -equivariant Fredholm operator between Hilbert spaces carrying unitary S^1 -actions. Its equivariant index is

$$\text{Ind}_{S^1}(T) := [\ker T] - [\ker T^*] \in R(S^1).$$

Define the interval operator

$$B := \partial_u + A_u$$

on

$$L^2([0, 1], \mathcal{H}) \cong L^2([0, 1] \times S^1, \mathbb{C}^m)$$

with domain

$$\text{Dom}(B) := \{f \in H^1([0, 1], \mathcal{H}) \cap L^2([0, 1], \mathcal{D}) : P_{>0}(A_0)f(0) = 0, \quad P_{<0}(A_1)f(1) = 0\}.$$

5.2 Fourier decomposition

For $n \in \mathbb{Z}$, set

$$e_n(\theta) = \frac{1}{\sqrt{2\pi}} e^{in\theta}, \quad \chi_n(e^{i\varphi}) = e^{in\varphi}.$$

Then

$$L^2(S^1) = \widehat{\bigoplus_{n \in \mathbb{Z}} \mathbb{C} e_n}, \quad -i \frac{d}{d\theta} e_n = n e_n.$$

Hence

$$\mathcal{H} \cong \widehat{\bigoplus_{n \in \mathbb{Z}} \chi_n \otimes \mathbb{C}^m},$$

and the restriction of A_u to the n -th sector is

$$a_n(u) = \text{diag}(n + \alpha_1(u), \dots, n + \alpha_{r_T}(u), n + \beta_1(u), \dots, n + \beta_{r_N}(u)).$$

Lemma 5.3. *With the APS boundary conditions determined by A_0 and A_1 , one has*

$$\text{Dom}(B) \cong \widehat{\bigoplus_{n \in \mathbb{Z}} \text{Dom}(B_n) \otimes \chi_n},$$

where

$$B_n = \bigoplus_{j=1}^{r_T} b_{n+\alpha_j} \oplus \bigoplus_{k=1}^{r_N} b_{n+\beta_k}.$$

Consequently,

$$B \cong \widehat{\bigoplus_{n \in \mathbb{Z}} B_n \otimes 1_{\chi_n}}.$$

Proof. The Fourier decomposition diagonalises $-i\partial_\theta$, hence diagonalises both endpoint operators A_0 and A_1 . Their spectral projections preserve every Fourier mode, so the APS boundary conditions decouple sectorwise. $\square$

5.3 Uniform invertibility outside a finite set

Lemma 5.4. *Let $c \in C([0, 1], \mathbb{R})$. If either*

$$c(u) \geq \delta > 0 \quad (\forall u \in [0, 1]),$$

or

$$c(u) \leq -\delta < 0 \quad (\forall u \in [0, 1]),$$

then b_c is invertible and

$$\|b_c^{-1}\|_{L^2 \rightarrow L^2} \leq \delta^{-1}.$$

Proof. If $c \geq \delta > 0$, the APS condition is $f(0) = 0$ and the unique solution of

$$f'(u) + c(u)f(u) = g(u)$$

is

$$f(u) = \int_0^u e^{-\int_s^u c(r) dr} g(s) ds.$$

Since $e^{-\int_s^u c(r) dr} \leq e^{-\delta(u-s)}$, Young's inequality gives

$$\|f\|_{L^2} \leq \delta^{-1} \|g\|_{L^2}.$$

The case $c \leq -\delta$ is identical with the boundary condition $f(1) = 0$. □

Lemma 5.5. *Let*

$$T_n : \text{Dom}(T_n) \subset H_n \rightarrow K_n$$

be a sequence of closed densely defined bijective operators between Hilbert spaces. Assume that each inverse

$$T_n^{-1} : K_n \rightarrow H_n$$

is bounded and that

$$\sup_n \|T_n^{-1}\|_{K_n \rightarrow H_n} < \infty.$$

Then the Hilbert direct sum

$$\bigoplus_n T_n : \text{Dom}\left(\bigoplus_n T_n\right) \subset \bigoplus_n H_n \rightarrow \bigoplus_n K_n$$

is bijective, with inverse

$$\left(\bigoplus_n T_n\right)^{-1} = \bigoplus_n T_n^{-1}.$$

In particular, the direct sum operator is closed and invertible.

Proof. Let $y = (y_n)_n \in \bigoplus_n K_n$ and set $x_n := T_n^{-1}y_n$. Then

$$\sum_n \|x_n\|_{H_n}^2 \leq \left(\sup_n \|T_n^{-1}\|_{K_n \rightarrow H_n}\right)^2 \sum_n \|y_n\|_{K_n}^2 < \infty,$$

so $x = (x_n)_n \in \bigoplus_n H_n$. Moreover

$$\sum_n \|T_n x_n\|_{K_n}^2 = \sum_n \|y_n\|_{K_n}^2 < \infty,$$

so $x \in \text{Dom}(\bigoplus_n T_n)$ and $(\bigoplus_n T_n)x = y$. Injectivity is immediate from injectivity of each T_n . □

5.4 The representation-valued index formula

Theorem 5.6. *The operator $B = \partial_u + A_u$ is S^1 -Fredholm, and its equivariant index is*

$$\text{Ind}_{S^1}(B) = \sum_{n \in \mathbb{Z}} \left(\sum_{j=1}^{r_T} \sigma_{n,j}^T + \sum_{k=1}^{r_N} \sigma_{n,k}^N \right) [\chi_n],$$

where

$$\begin{aligned} \sigma_{n,j}^T &= \frac{1}{2} \left(\text{sgn}(n + \alpha_j(1)) - \text{sgn}(n + \alpha_j(0)) \right), \\ \sigma_{n,k}^N &= \frac{1}{2} \left(\text{sgn}(n + \beta_k(1)) - \text{sgn}(n + \beta_k(0)) \right). \end{aligned}$$

Moreover only finitely many Fourier modes contribute nontrivially.

Proof. Set

$$M_0 := 1 + \left\lceil \max \left\{ \max_j \|\alpha_j\|_{L^\infty([0,1])}, \max_k \|\beta_k\|_{L^\infty([0,1])} \right\} \right\rceil.$$

If $|n| \geq M_0$, then for all $u \in [0, 1]$,

$$|n + \alpha_j(u)| \geq 1, \quad |n + \beta_k(u)| \geq 1.$$

Hence each scalar coefficient has a fixed sign on the whole interval. By Lemma 5.4, every scalar summand is invertible with inverse norm bounded by 1. Therefore each B_n is invertible, and $\|B_n^{-1}\|$ is uniformly bounded for all $|n| \geq M_0$.

Let

$$F := \{n \in \mathbb{Z} : |n| < M_0\}.$$

Then F is finite and

$$B = B_F \oplus B_{F^c},$$

with

$$B_F = \bigoplus_{n \in F} B_n \otimes 1_{\chi_n}, \quad B_{F^c} = \bigoplus_{n \notin F} B_n \otimes 1_{\chi_n}.$$

By Lemma 5.5, B_{F^c} is invertible. Since B_F is a finite direct sum of Fredholm operators, it is Fredholm. Hence B is S^1 -Fredholm.

For each fixed n , Theorem 4.4 gives

$$\text{ind}(b_{n+\alpha_j}) = \frac{1}{2} \left(\text{sgn}(n + \alpha_j(1)) - \text{sgn}(n + \alpha_j(0)) \right) = \sigma_{n,j}^T,$$

and similarly

$$\text{ind}(b_{n+\beta_k}) = \sigma_{n,k}^N.$$

Hence

$$\text{ind}(B_n) = \sum_{j=1}^{r_T} \sigma_{n,j}^T + \sum_{k=1}^{r_N} \sigma_{n,k}^N.$$

Only finitely many modes contribute, so

$$\text{Ind}_{S^1}(B) = \sum_{n \in \mathbb{Z}} \text{ind}(B_n) [\chi_n].$$

□

Corollary 5.7. *Assume that for every nuisance channel k ,*

$$-\beta_k([0, 1]) \cap \mathbb{Z} = \emptyset.$$

Then

$$\sigma_{n,k}^N = 0 \quad (\forall n \in \mathbb{Z}, \forall k),$$

and hence

$$\text{Ind}_{S^1}(B) = \sum_{n \in \mathbb{Z}} \left(\sum_{j=1}^{r_T} \sigma_{n,j}^T \right) [\chi_n].$$

Proof. Fix $n \in \mathbb{Z}$ and k . Since $-\beta_k([0, 1]) \cap \mathbb{Z} = \emptyset$, the continuous function

$$u \mapsto n + \beta_k(u)$$

never vanishes on $[0, 1]$. Hence its endpoint values have the same sign, so

$$\sigma_{n,k}^N = 0.$$

Substitute into Theorem 5.6. □

Corollary 5.8. *If a task channel α_j is monotone increasing, then*

$$\sigma_{n,j}^T = \begin{cases} 1, & -\alpha_j(1) < n < -\alpha_j(0), \\ 0, & \text{otherwise.} \end{cases}$$

If α_j is monotone decreasing, then

$$\sigma_{n,j}^T = \begin{cases} -1, & -\alpha_j(0) < n < -\alpha_j(1), \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If α_j is increasing, then $\sigma_{n,j}^T = 1$ exactly when

$$n + \alpha_j(0) < 0 < n + \alpha_j(1),$$

which is equivalent to $-\alpha_j(1) < n < -\alpha_j(0)$. The decreasing case is identical. □

6 A continuous-spectrum Plancherel toy model

This section gives a continuous-spectrum model in which the selective spectral-flow density can be computed explicitly. The only external theorem used here is the semifinite spectral-flow=APS-index result of Azzali–Wahl.

6.1 The semifinite von Neumann algebra

Let

$$m := r_T + r_N, \quad \mathcal{H}_{\text{Pl}} := L^2(\mathbb{R}, \mathbb{C}^m).$$

Define

$$\mathcal{N}_{\text{Pl}} := L^\infty(\mathbb{R}, M_m(\mathbb{C}))$$

acting by multiplication,

$$(M_{q(\lambda)} f)(\lambda) = q(\lambda) f(\lambda).$$

For a positive multiplier $M_{q(\lambda)} \in \mathcal{N}_{\text{Pl}}$, define

$$\text{Tr}_{\text{Pl}}(M_{q(\lambda)}) = \int_{\mathbb{R}} \text{tr}(q(\lambda)) d\lambda.$$

Let $K(\mathcal{N}_{\text{Pl}})$ be the norm-closed two-sided ideal generated by the finite-trace projections, and define

$$\mathcal{M}_{\text{APS}} := B(L^2([0, 1])) \bar{\otimes} \mathcal{N}_{\text{Pl}}, \quad \widetilde{\text{Tr}} := \text{Tr}_{B(L^2([0, 1]))} \bar{\otimes} \text{Tr}_{\text{Pl}}.$$

Lemma 6.1. *For a multiplier $T = M_{q(\lambda)} \in \mathcal{N}_{\text{Pl}}$, the following are equivalent:*

1. $T \in K(\mathcal{N}_{\text{Pl}})$;
2. for every $\varepsilon > 0$, the set

$$E_\varepsilon(T) := \{\lambda \in \mathbb{R} : \|q(\lambda)\| \geq \varepsilon\}$$

has finite Lebesgue measure.

Proof. If $P = M_{p(\lambda)}$ is a finite-trace projection, then

$$\text{Tr}_{\text{Pl}}(P) = \int_{\mathbb{R}} \text{rank } p(\lambda) d\lambda < \infty.$$

Since $\text{rank } p(\lambda) \geq 1$ on $\text{supp } P$, one gets

$$\mu(\text{supp } P) \leq \text{Tr}_{\text{Pl}}(P) < \infty.$$

Hence every finite-trace projection satisfies condition (2), and therefore so does every element of the ideal generated by them. The set of multipliers satisfying (2) is a closed two-sided ideal, so

$$K(\mathcal{N}_{\text{Pl}}) \subset \{T : \text{condition (2) holds}\}.$$

Conversely, assume condition (2). Set

$$E_n := \{\lambda : \|q(\lambda)\| \geq 1/n\}, \quad P_n := M_{1_{E_n}} I.$$

Then $\mu(E_n) < \infty$, so P_n is a finite-trace projection and therefore belongs to $K(\mathcal{N}_{\text{Pl}})$. Moreover,

$$\|T - TP_n\| = \text{ess sup}_{\lambda \notin E_n} \|q(\lambda)\| \leq 1/n.$$

Since $TP_n \in K(\mathcal{N}_{\text{Pl}})$ and $K(\mathcal{N}_{\text{Pl}})$ is norm-closed, we obtain $T \in K(\mathcal{N}_{\text{Pl}})$. □

6.2 The multiplier path

Let

$$\alpha_j \in C^1([0, 1], \mathbb{R}), \quad j = 1, \dots, r_T,$$

and assume each α_j is constant near $u = 0$ and near $u = 1$.

Let

$$\gamma_k \in C([0, 1], [\gamma_0, \infty)), \quad \gamma_0 > 0, \quad k = 1, \dots, r_N,$$

and assume each γ_k is also constant near the endpoints.

Define, for each $(u, \lambda) \in [0, 1] \times \mathbb{R}$,

$$a_u(\lambda) = \text{diag}\left(\lambda + \alpha_1(u), \dots, \lambda + \alpha_{r_T}(u), \sqrt{\lambda^2 + \gamma_1(u)^2}, \dots, \sqrt{\lambda^2 + \gamma_{r_N}(u)^2}\right),$$

and define

$$A_u := M_{a_u(\lambda)}.$$

Set

$$\mathcal{D} := \left\{ f \in \mathcal{H}_{\text{Pl}} : (1 + \lambda^2)^{1/2} f(\lambda) \in L^2(\mathbb{R}, \mathbb{C}^m) \right\}.$$

Lemma 6.2. *Each A_u is a selfadjoint multiplier operator affiliated to $\mathcal{N}_{\text{Pl}}$, with common domain $\mathcal{D}$. Moreover $A_u - A_v$ is bounded on $\mathcal{H}_{\text{Pl}}$ for all $u, v \in [0, 1]$, and the path $u \mapsto A_u$ is continuous as a bounded path $H(A_0) \rightarrow \mathcal{H}_{\text{Pl}}$.*

Proof. Let

$$M_\lambda := M_{\lambda I_m}$$

be multiplication by the scalar variable λ . Its domain is exactly $\mathcal{D}$. For the task channels, multiplication by $\lambda + \alpha_j(u)$ differs from M_λ by the bounded scalar $\alpha_j(u)$, so the domain is still $\mathcal{D}$. For the nuisance channels, define

$$\eta_{k,u}(\lambda) := \sqrt{\lambda^2 + \gamma_k(u)^2} - |\lambda|.$$

Then

$$0 \leq \eta_{k,u}(\lambda) \leq \gamma_k(u) \leq \|\gamma_k\|_{L^\infty([0,1])}.$$

Hence multiplication by $\sqrt{\lambda^2 + \gamma_k(u)^2}$ differs from multiplication by $|\lambda|$ by a bounded operator. Since multiplication by $|\lambda|$ has the same domain as multiplication by λ , namely $\mathcal{D}$, the nuisance channel domains are also $\mathcal{D}$. Thus every A_u has common domain $\mathcal{D}$.

For the task channels,

$$(\lambda + \alpha_j(u)) - (\lambda + \alpha_j(v)) = \alpha_j(u) - \alpha_j(v),$$

which is bounded in λ . For the nuisance channels, the map $x \mapsto \sqrt{\lambda^2 + x^2}$ is 1-Lipschitz for $x \geq 0$, so

$$\left| \sqrt{\lambda^2 + \gamma_k(u)^2} - \sqrt{\lambda^2 + \gamma_k(v)^2} \right| \leq |\gamma_k(u) - \gamma_k(v)|$$

uniformly in λ . Hence $A_u - A_v$ is bounded. $\square$

Proposition 6.3. *For every $u \in [0, 1]$,*

$$(A_u \pm i)^{-1} \in K(\mathcal{N}_{\text{Pl}}).$$

Proof. Since A_u is a multiplier operator,

$$(A_u \pm i)^{-1} = M_{(a_u(\lambda) \pm i)^{-1}}.$$

By Lemma 6.1, it suffices to show that for each $\varepsilon > 0$,

$$E_\varepsilon(u) := \{ \lambda \in \mathbb{R} : \|(a_u(\lambda) \pm i)^{-1}\| \geq \varepsilon \}$$

has finite measure.

For the j -th task channel,

$$|(\lambda + \alpha_j(u) \pm i)^{-1}| = \frac{1}{\sqrt{(\lambda + \alpha_j(u))^2 + 1}}.$$

Hence

$$|(\lambda + \alpha_j(u) \pm i)^{-1}| \geq \varepsilon \implies (\lambda + \alpha_j(u))^2 \leq \varepsilon^{-2} - 1,$$

which confines λ to a bounded interval.

For the k -th nuisance channel,

$$\left| (\sqrt{\lambda^2 + \gamma_k(u)^2} \pm i)^{-1} \right| = \frac{1}{\sqrt{\lambda^2 + \gamma_k(u)^2 + 1}},$$

so the same conclusion follows immediately. Thus every $E_\varepsilon(u)$ has finite measure. $\square$

6.3 The global APS operator

Set

$$E_{\text{Pl}} := \{-\alpha_j(0), -\alpha_j(1) : 1 \leq j \leq r_T\}.$$

This is a finite set, hence of measure zero.

For each $\lambda \notin E_{\text{Pl}}$, define the fiberwise APS operator

$$D_{\lambda, \text{APS}} := \frac{d}{du} + a_u(\lambda)$$

on $L^2([0, 1], \mathbb{C}^m)$, with APS boundary conditions determined by the endpoint matrices $a_0(\lambda)$ and $a_1(\lambda)$.

Define the global APS operator as the direct integral

$$D_{\text{APS}} := \int_{\mathbb{R}}^{\oplus} D_{\lambda, \text{APS}} d\lambda,$$

with domain

$$\text{Dom}(D_{\text{APS}}) := \left\{ f \in \int_{\mathbb{R}}^{\oplus} \text{Dom}(D_{\lambda, \text{APS}}) d\lambda : \int_{\mathbb{R}} \|D_{\lambda, \text{APS}} f(\lambda)\|_{L^2([0, 1], \mathbb{C}^m)}^2 d\lambda < \infty \right\}.$$

Remark 6.4. At the exceptional points $\lambda \in E_{\text{Pl}}$, one has endpoint resonance in some task channel. Since E_{Pl} has measure zero, any measurable choice of boundary condition at those points leads to the same Breuer index.

Lemma 6.5. *The task spectral-flow density*

$$\nu_T(\lambda) = \sum_{j=1}^{r_T} \frac{1}{2} \left(\text{sgn}(\lambda + \alpha_j(1)) - \text{sgn}(\lambda + \alpha_j(0)) \right)$$

belongs to $L^1(\mathbb{R})$.

Proof. For each fixed j , the function

$$\lambda \mapsto \frac{1}{2} \left(\text{sgn}(\lambda + \alpha_j(1)) - \text{sgn}(\lambda + \alpha_j(0)) \right)$$

is either the indicator function of a bounded interval, minus such an indicator, or zero. Hence it lies in $L^1(\mathbb{R})$. Summing over finitely many channels preserves integrability. $\square$

Lemma 6.6. *One has*

$$\ker D_{\text{APS}} = \int_{\mathbb{R}}^{\oplus} \ker D_{\lambda, \text{APS}} d\lambda, \quad \ker D_{\text{APS}}^* = \int_{\mathbb{R}}^{\oplus} \ker D_{\lambda, \text{APS}}^* d\lambda.$$

Proof. By definition of the direct-integral operator,

$$f \in \ker D_{\text{APS}} \iff f(\lambda) \in \text{Dom}(D_{\lambda, \text{APS}}) \text{ for a.e. } \lambda \quad \text{and} \quad D_{\lambda, \text{APS}} f(\lambda) = 0 \text{ for a.e. } \lambda.$$

This is exactly the direct integral of the fiber kernels. The adjoint statement is identical. $\square$

Lemma 6.7. *Fix a task channel j and define*

$$I_j^+ := \{\lambda \in \mathbb{R} : \lambda + \alpha_j(0) < 0 < \lambda + \alpha_j(1)\}.$$

For $\lambda \in I_j^+$, define

$$\psi_{j, \lambda}(u) := \exp\left(-\int_0^u (\lambda + \alpha_j(s)) ds\right) \in L^2([0, 1]).$$

Then the map $\lambda \mapsto \psi_{j,\lambda}$ is continuous from I_j^+ into $L^2([0, 1])$. In particular, the normalized vectors

$$\phi_{j,\lambda} := \frac{\psi_{j,\lambda}}{\|\psi_{j,\lambda}\|_{L^2([0,1])}}$$

form a measurable unit vector field on I_j^+ , and the rank-one projections

$$p_j^+(\lambda) := |\phi_{j,\lambda}\rangle\langle\phi_{j,\lambda}|$$

define a measurable projection field. The same holds for the adjoint field on

$$I_j^- := \{\lambda \in \mathbb{R} : \lambda + \alpha_j(0) > 0 > \lambda + \alpha_j(1)\}.$$

Proof. For fixed $u \in [0, 1]$,

$$\lambda \mapsto \psi_{j,\lambda}(u) = e^{-\lambda u} \exp\left(-\int_0^u \alpha_j(s) ds\right)$$

is continuous. On every compact subinterval of I_j^+ , these functions are dominated by an $L^2([0, 1])$ -function independent of λ . Dominated convergence therefore gives continuity in $L^2([0, 1])$. Normalization preserves measurability, and so does passage to the rank-one projection. The adjoint case is identical. $\square$

Proposition 6.8. *For the global APS operator D_{APS} one has:*

1. *in the j -th task channel,*

$$\widetilde{\text{Tr}}(P_{\ker D_{\text{APS},j}}) = |I_j^+|;$$

2. *in the j -th task channel,*

$$\widetilde{\text{Tr}}(P_{\ker D_{\text{APS},j}^*}) = |I_j^-|;$$

3. *every nuisance channel contributes trivially to both kernels.*

Consequently, whenever D_{APS} is Breuer–Fredholm,

$$\text{Ind}_{\widetilde{\text{Tr}}}(D_{\text{APS}}) = \sum_{j=1}^{r_{\text{T}}} (|I_j^+| - |I_j^-|) = \sum_{j=1}^{r_{\text{T}}} (\alpha_j(1) - \alpha_j(0)).$$

Proof. For $\lambda \notin E_{\text{P1}}$, the scalar fiber operator in the j -th task channel is

$$b_{\lambda+\alpha_j} = \frac{d}{du} + (\lambda + \alpha_j(u)).$$

By Corollary 4.6,

$$\dim \ker b_{\lambda+\alpha_j} = 1 \iff \lambda \in I_j^+, \quad \dim \ker b_{\lambda+\alpha_j}^* = 1 \iff \lambda \in I_j^-.$$

By Lemma 6.7, the corresponding kernel projection fields are measurable rank-one fields on $I_j^\pm$. Hence, on the j -th task channel,

$$P_{\ker D_{\text{APS},j}} = \int_{\mathbb{R}}^{\oplus} 1_{I_j^+}(\lambda) p_j^+(\lambda) d\lambda, \quad P_{\ker D_{\text{APS},j}^*} = \int_{\mathbb{R}}^{\oplus} 1_{I_j^-}(\lambda) p_j^-(\lambda) d\lambda.$$

Since the fiber rank is 1 on $I_j^\pm$ and 0 elsewhere,

$$\widetilde{\text{Tr}}(P_{\ker D_{\text{APS},j}}) = |I_j^+|, \quad \widetilde{\text{Tr}}(P_{\ker D_{\text{APS},j}^*}) = |I_j^-|.$$

For a nuisance channel k , the scalar coefficient is

$$u \mapsto \sqrt{\lambda^2 + \gamma_k(u)^2},$$

which is strictly positive for all (u, λ) . Hence the corresponding scalar APS kernel and adjoint kernel vanish identically. Therefore

$$\widetilde{\text{Tr}}(P_{\ker D_{\text{APS}}}) - \widetilde{\text{Tr}}(P_{\ker D_{\text{APS}}^*}) = \sum_{j=1}^{r_T} (|I_j^+| - |I_j^-|).$$

Finally, for each j one has $|I_j^+| - |I_j^-| = \alpha_j(1) - \alpha_j(0)$. $\square$

Theorem 6.9. *The path $(A_u)_{u \in [0,1]}$ satisfies the hypotheses of the Azzali–Wahl APS theorem. Hence the semifinite spectral flow is well-defined and*

$$\text{sf}_{\text{Tr}_{\text{Pl}}} \{A_u\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}}}(D_{\text{APS}}) = \int_{\mathbb{R}} \nu_T(\lambda) d\lambda = \sum_{j=1}^{r_T} (\alpha_j(1) - \alpha_j(0)).$$

In particular, the nuisance channels make no contribution.

Proof. By Lemma 6.2, the path has common domain and is continuous in graph norm. By Proposition 6.3,

$$(A_u \pm i)^{-1} \in K(\mathcal{N}_{\text{Pl}}) \quad (\forall u \in [0, 1]).$$

Since the path is constant near the endpoints, the APS theorem of Azzali–Wahl applies and yields

$$\text{sf}_{\text{Tr}_{\text{Pl}}} \{A_u\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}}}(D_{\text{APS}}).$$

Now Proposition 6.8 gives

$$\text{Ind}_{\widetilde{\text{Tr}}}(D_{\text{APS}}) = \sum_{j=1}^{r_T} (\alpha_j(1) - \alpha_j(0)).$$

Finally, by Lemma 6.5 and the elementary interval computation,

$$\int_{\mathbb{R}} \nu_T(\lambda) d\lambda = \sum_{j=1}^{r_T} (\alpha_j(1) - \alpha_j(0)).$$

Combining the displayed equalities proves the theorem. $\square$

Corollary 6.10. *If the task channels are also replaced by uniformly positive gap channels, so that their fiber coefficients never cross zero for any (u, λ) , then*

$$\text{sf}_{\text{Tr}_{\text{Pl}}} \{A_u\}_{u \in [0,1]} = 0.$$

Proof. Under the stated hypothesis, every task fiber has zero scalar APS index, just as in the nuisance case. Hence the right-hand side of Theorem 6.9 vanishes. $\square$

7 A genuinely structured compact model on the sphere

In this section we pass from the abelian compact model of Section 5 to a nonabelian compact homogeneous space. The resulting theorem is fully rigorous and self-contained.

7.1 The spherical harmonic decomposition

Let

$$K = SO(3), \quad \Sigma = S^2.$$

The unitary $SO(3)$ -representation on $L^2(S^2)$ decomposes as

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty} V_{\ell}},$$

where V_{ℓ} is the irreducible representation of highest weight ℓ , of dimension

$$\dim V_{\ell} = 2\ell + 1.$$

Let

$$r_T, r_N \in \mathbb{N} \cup \{0\}, \quad m := r_T + r_N.$$

Set

$$\mathcal{H}_{S^2} := L^2(S^2, \mathbb{C}^m) \cong \widehat{\bigoplus_{\ell=0}^{\infty} V_{\ell} \otimes \mathbb{C}^m}.$$

7.2 Spectral-symbol data

Let

$$(p_{j,\ell})_{\ell \geq 0} \subset \mathbb{R}, \quad 1 \leq j \leq r_T,$$

and

$$(q_{k,\ell})_{\ell \geq 0} \subset \mathbb{R}, \quad 1 \leq k \leq r_N,$$

be real sequences.

Let

$$\alpha_j, \beta_k \in C^1([0, 1], \mathbb{R}),$$

and assume each is constant near $u = 0$ and near $u = 1$.

For every $\ell \geq 0$, define

$$a_{\ell}(u) = \text{diag}(p_{1,\ell} + \alpha_1(u), \dots, p_{r_T,\ell} + \alpha_{r_T}(u), q_{1,\ell} + \beta_1(u), \dots, q_{r_N,\ell} + \beta_{r_N}(u)).$$

Define the reference operator A_0 by

$$A_0|_{V_{\ell} \otimes \mathbb{C}^m} = 1_{V_{\ell}} \otimes a_{\ell}(0),$$

with domain

$$\mathcal{D}_{S^2} := D(A_0) = \left\{ f = (f_{\ell})_{\ell \geq 0} : \sum_{\ell=0}^{\infty} \|a_{\ell}(0) f_{\ell}\|^2 < \infty \right\}.$$

For each $u \in [0, 1]$, define

$$A_u : \mathcal{D}_{S^2} \rightarrow \mathcal{H}_{S^2}, \quad A_u|_{V_{\ell} \otimes \mathbb{C}^m} = 1_{V_{\ell}} \otimes a_{\ell}(u).$$

Lemma 7.1. *Each A_u is selfadjoint on the common domain $\mathcal{D}_{S^2}$. Moreover, for all $u, v \in [0, 1]$, the difference $A_u - A_v$ is bounded on $\mathcal{H}_{S^2}$.*

Proof. The operator A_0 is the orthogonal direct sum of finite-dimensional selfadjoint operators, hence is selfadjoint on its natural diagonal domain $D(A_0) = \mathcal{D}_{S^2}$. For every u one has

$$A_u = A_0 + B_u,$$

where

$$B_u|_{V_{\ell} \otimes \mathbb{C}^m} = 1_{V_{\ell}} \otimes (a_{\ell}(u) - a_{\ell}(0)).$$

But

$$a_\ell(u) - a_\ell(0) = \text{diag}(\alpha_1(u) - \alpha_1(0), \dots, \alpha_{r_T}(u) - \alpha_{r_T}(0), \beta_1(u) - \beta_1(0), \dots, \beta_{r_N}(u) - \beta_{r_N}(0)),$$

which is independent of ℓ . Hence B_u is bounded on $\mathcal{H}_{S^2}$. By the bounded perturbation theorem, A_u is selfadjoint on $D(A_0) = \mathcal{D}_{S^2}$. The same argument with u and v gives boundedness of $A_u - A_v$. $\square$

Definition 7.2. If T is an $SO(3)$ -equivariant Fredholm operator, its equivariant index is

$$\text{Ind}_{SO(3)}(T) := [\ker T] - [\ker T^*] \in R(SO(3)).$$

Assume the endpoint nonresonance condition: for all $\ell \geq 0$,

$$p_{j,\ell} + \alpha_j(0) \neq 0, \quad p_{j,\ell} + \alpha_j(1) \neq 0 \quad (1 \leq j \leq r_T),$$

and

$$q_{k,\ell} + \beta_k(0) \neq 0, \quad q_{k,\ell} + \beta_k(1) \neq 0 \quad (1 \leq k \leq r_N).$$

Define

$$B_{S^2} := \partial_u + A_u$$

on

$$L^2([0, 1], \mathcal{H}_{S^2})$$

with domain

$$\text{Dom}(B_{S^2}) := \{f \in H^1([0, 1], \mathcal{H}_{S^2}) \cap L^2([0, 1], \mathcal{D}_{S^2}) : P_{>0}(A_0)f(0) = 0, P_{<0}(A_1)f(1) = 0\}.$$

Lemma 7.3. One has

$$\text{Dom}(B_{S^2}) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} \text{Dom}(B_\ell) \otimes V_\ell,$$

where

$$B_\ell = \bigoplus_{j=1}^{r_T} b_{p_{j,\ell} + \alpha_j} \oplus \bigoplus_{k=1}^{r_N} b_{q_{k,\ell} + \beta_k}.$$

Consequently,

$$B_{S^2} \cong \widehat{\bigoplus_{\ell=0}^{\infty}} B_\ell \otimes 1_{V_\ell}.$$

Proof. The endpoint operators A_0 and A_1 are diagonal on the spherical harmonic decomposition

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell.$$

Hence their spectral projections preserve every ℓ -sector, and the APS boundary conditions decouple sectorwise. $\square$

Assume there exist $L_0 \in \mathbb{N}$ and $\delta > 0$ such that for all $\ell \geq L_0$ and all $u \in [0, 1]$,

$$|p_{j,\ell} + \alpha_j(u)| \geq \delta \quad (1 \leq j \leq r_T),$$

and

$$|q_{k,\ell} + \beta_k(u)| \geq \delta \quad (1 \leq k \leq r_N).$$

Theorem 7.4. *Under the assumptions above, B_{S^2} is $SO(3)$ -Fredholm and*

$$\text{Ind}_{SO(3)}(B_{S^2}) = \sum_{\ell=0}^{\infty} \left(\sum_{j=1}^{r_T} \sigma_{\ell,j}^T + \sum_{k=1}^{r_N} \sigma_{\ell,k}^N \right) [V_\ell],$$

where

$$\begin{aligned} \sigma_{\ell,j}^T &= \frac{1}{2} \left(\text{sgn}(p_{j,\ell} + \alpha_j(1)) - \text{sgn}(p_{j,\ell} + \alpha_j(0)) \right), \\ \sigma_{\ell,k}^N &= \frac{1}{2} \left(\text{sgn}(q_{k,\ell} + \beta_k(1)) - \text{sgn}(q_{k,\ell} + \beta_k(0)) \right). \end{aligned}$$

Moreover only finitely many ℓ contribute nontrivially.

Proof. For each fixed ℓ , the sector operator

$$B_\ell = \bigoplus_{j=1}^{r_T} b_{p_{j,\ell} + \alpha_j} \oplus \bigoplus_{k=1}^{r_N} b_{q_{k,\ell} + \beta_k}$$

is a finite direct sum of scalar APS operators of the type studied in Section 4. If $\ell \geq L_0$, then every scalar coefficient has a uniform gap of size at least δ from zero on the whole interval. Hence, by Lemma 5.4, every scalar summand is invertible with inverse norm at most δ^{-1} , and therefore B_ℓ is invertible with uniformly bounded inverse.

Thus the tail

$$\widehat{\bigoplus_{\ell \geq L_0} B_\ell \otimes 1_{V_\ell}}$$

is invertible by Lemma 5.5, while the finite head

$$\bigoplus_{\ell=0}^{L_0-1} B_\ell \otimes 1_{V_\ell}$$

is Fredholm. Hence B_{S^2} is $SO(3)$ -Fredholm.

For each fixed ℓ , Theorem 4.4 gives

$$\text{ind}(b_{p_{j,\ell} + \alpha_j}) = \frac{1}{2} \left(\text{sgn}(p_{j,\ell} + \alpha_j(1)) - \text{sgn}(p_{j,\ell} + \alpha_j(0)) \right) = \sigma_{\ell,j}^T,$$

and similarly

$$\text{ind}(b_{q_{k,\ell} + \beta_k}) = \sigma_{\ell,k}^N.$$

Therefore

$$\text{ind}(B_\ell) = \sum_{j=1}^{r_T} \sigma_{\ell,j}^T + \sum_{k=1}^{r_N} \sigma_{\ell,k}^N.$$

Since the ℓ -th sector carries the irreducible representation V_ℓ ,

$$\text{Ind}_{SO(3)}(B_{S^2}) = \sum_{\ell=0}^{\infty} \text{ind}(B_\ell) [V_\ell].$$

Only finitely many sectors are nonzero because all sectors with $\ell \geq L_0$ are invertible. □

Corollary 7.5. *If, in addition, for every nuisance channel k and every $\ell \geq 0$,*

$$-q_{k,\ell} \notin \beta_k([0, 1]),$$

then

$$\sigma_{\ell,k}^N = 0 \quad (\forall \ell \geq 0, \forall k),$$

and hence

$$\text{Ind}_{SO(3)}(B_{S^2}) = \sum_{\ell=0}^{\infty} \left(\sum_{j=1}^{r_T} \sigma_{\ell,j}^T \right) [V_\ell].$$

Proof. Fix ℓ and k . The condition $-q_{k,\ell} \notin \beta_k([0, 1])$ means that the continuous function $u \mapsto q_{k,\ell} + \beta_k(u)$ never vanishes on $[0, 1]$. Hence its endpoint values have the same sign, so $\sigma_{\ell,k}^N = 0$. Substitute into Theorem 7.4. $\square$

Corollary 7.6. *Let*

$$-\Delta_{S^2}|_{V_\ell} = \ell(\ell + 1).$$

Fix real functions

$$\varphi_j, \psi_k : [0, \infty) \rightarrow \mathbb{R}$$

and define

$$p_{j,\ell} := \varphi_j(\ell(\ell + 1)), \quad q_{k,\ell} := \psi_k(\ell(\ell + 1)).$$

Assume that the endpoint nonresonance and high-frequency gap hypotheses of Theorem 7.4 are satisfied for these choices. Then the theorem applies to the geometric spectral-filter family

$$A_u^{sph} = \bigoplus_{j=1}^{r_T} (\alpha_j(u)I + \varphi_j(-\Delta_{S^2})) \oplus \bigoplus_{k=1}^{r_N} (\beta_k(u)I + \psi_k(-\Delta_{S^2})),$$

and the corresponding equivariant index is

$$\text{Ind}_{SO(3)}(B_{S^2}^{sph}) = \sum_{\ell=0}^{\infty} \left(\sum_{j=1}^{r_T} \sigma_{\ell,j}^T + \sum_{k=1}^{r_N} \sigma_{\ell,k}^N \right) [V_\ell],$$

with

$$\begin{aligned} \sigma_{\ell,j}^T &= \frac{1}{2} \left(\text{sgn}(\varphi_j(\ell(\ell + 1)) + \alpha_j(1)) - \text{sgn}(\varphi_j(\ell(\ell + 1)) + \alpha_j(0)) \right), \\ \sigma_{\ell,k}^N &= \frac{1}{2} \left(\text{sgn}(\psi_k(\ell(\ell + 1)) + \beta_k(1)) - \text{sgn}(\psi_k(\ell(\ell + 1)) + \beta_k(0)) \right). \end{aligned}$$

Proof. This is exactly Theorem 7.4 with the choices

$$p_{j,\ell} = \varphi_j(\ell(\ell + 1)), \quad q_{k,\ell} = \psi_k(\ell(\ell + 1)).$$

$\square$

8 Weighted Plancherel models and a radial hyperbolic application

This section first upgrades the continuous-spectrum toy model of Section 6 to a weighted spectral measure, and then applies that result to a genuinely geometric radial model on real hyperbolic space.

8.1 A weighted-measure version of the continuous-spectrum model

Let μ be a positive Borel measure on $\mathbb{R}$ such that every bounded interval has finite μ -measure. Set

$$\mathcal{H}_\mu := L^2(\mathbb{R}, \mathbb{C}^m; d\mu), \quad \mathcal{N}_\mu := L^\infty(\mathbb{R}, \mu; M_m(\mathbb{C})),$$

acting by multiplication on $\mathcal{H}_\mu$. For a positive multiplier $M_{q(\lambda)} \in \mathcal{N}_\mu$, define

$$\text{Tr}_\mu(M_{q(\lambda)}) = \int_{\mathbb{R}} \text{tr}(q(\lambda)) d\mu(\lambda).$$

Set

$$\mathcal{M}_{\mu, \text{APS}} := B(L^2([0, 1])) \bar{\otimes} \mathcal{N}_\mu, \quad \widetilde{\text{Tr}}_\mu := \text{Tr}_{B(L^2([0, 1]))} \bar{\otimes} \text{Tr}_\mu.$$

Let

$$\alpha_j \in C^1([0, 1], \mathbb{R}), \quad 1 \leq j \leq r_T,$$

be constant near the endpoints, and let

$$\gamma_k \in C([0, 1], [\gamma_0, \infty)), \quad \gamma_0 > 0, \quad 1 \leq k \leq r_N,$$

also be constant near the endpoints.

Define the exceptional set

$$E_\mu := \{-\alpha_j(0), -\alpha_j(1) : 1 \leq j \leq r_T\},$$

and assume

$$\mu(E_\mu) = 0.$$

Define

$$a_u^\mu(\lambda) = \text{diag}\left(\lambda + \alpha_1(u), \dots, \lambda + \alpha_{r_T}(u), \sqrt{\lambda^2 + \gamma_1(u)^2}, \dots, \sqrt{\lambda^2 + \gamma_{r_N}(u)^2}\right),$$

and

$$A_u^\mu := M_{a_u^\mu(\lambda)}.$$

For $\lambda \notin E_\mu$ define

$$\nu_T^\mu(\lambda) = \sum_{j=1}^{r_T} \frac{1}{2} \left(\text{sgn}(\lambda + \alpha_j(1)) - \text{sgn}(\lambda + \alpha_j(0)) \right),$$

and define it arbitrarily on the null set E_μ .

For each task channel j , define

$$I_j^+ := \{\lambda \in \mathbb{R} : \lambda + \alpha_j(0) < 0 < \lambda + \alpha_j(1)\}, \quad I_j^- := \{\lambda \in \mathbb{R} : \lambda + \alpha_j(0) > 0 > \lambda + \alpha_j(1)\}.$$

For $\lambda \notin E_\mu$, define the fiberwise APS operator

$$D_{\lambda, \text{APS}}^\mu := \frac{d}{du} + a_u^\mu(\lambda)$$

on $L^2([0, 1], \mathbb{C}^m)$ with APS boundary conditions determined by the endpoint matrices $a_0^\mu(\lambda)$ and $a_1^\mu(\lambda)$. Define the global APS operator

$$D_{\mu, \text{APS}} := \int_{\mathbb{R}}^\oplus D_{\lambda, \text{APS}}^\mu d\mu(\lambda),$$

with domain

$$\text{Dom}(D_{\mu, \text{APS}}) := \left\{ f \in \int_{\mathbb{R}}^\oplus \text{Dom}(D_{\lambda, \text{APS}}^\mu) d\mu(\lambda) : \int_{\mathbb{R}} \|D_{\lambda, \text{APS}}^\mu f(\lambda)\|_{L^2([0, 1], \mathbb{C}^m)}^2 d\mu(\lambda) < \infty \right\}.$$

Lemma 8.1. *For a multiplier $T = M_{q(\lambda)} \in \mathcal{N}_\mu$, the following are equivalent:*

1. $T \in K(\mathcal{N}_\mu)$;
2. for every $\varepsilon > 0$, the set

$$E_\varepsilon(T) := \{\lambda \in \mathbb{R} : \|q(\lambda)\| \geq \varepsilon\}$$

has finite μ -measure.

Proof. The proof is identical to that of Lemma 6.1, replacing Lebesgue measure by μ . The only input needed is that bounded intervals have finite μ -measure. $\square$

Lemma 8.2. *One has*

$$\ker D_{\mu, \text{APS}} = \int_{\mathbb{R}}^{\oplus} \ker D_{\lambda, \text{APS}}^{\mu} d\mu(\lambda), \quad \ker D_{\mu, \text{APS}}^* = \int_{\mathbb{R}}^{\oplus} \ker (D_{\lambda, \text{APS}}^{\mu})^* d\mu(\lambda).$$

Proof. This is the same direct-integral kernel argument as in Lemma 6.6, now with the measure μ . $\square$

Proposition 8.3. *For the global weighted APS operator $D_{\mu, \text{APS}}$ one has:*

1. *in the j -th task channel,*

$$\widetilde{\text{Tr}}_{\mu}(P_{\ker D_{\mu, \text{APS}, j}}) = \mu(I_j^+);$$

2. *in the j -th task channel,*

$$\widetilde{\text{Tr}}_{\mu}(P_{\ker D_{\mu, \text{APS}, j}^*}) = \mu(I_j^-);$$

3. *every nuisance channel contributes trivially to both $\ker D_{\mu, \text{APS}}$ and $\ker D_{\mu, \text{APS}}^*$.*

Consequently, whenever $D_{\mu, \text{APS}}$ is Breuer–Fredholm,

$$\text{Ind}_{\widetilde{\text{Tr}}_{\mu}}(D_{\mu, \text{APS}}) = \sum_{j=1}^{r_T} \mu(I_j^+) - \sum_{j=1}^{r_T} \mu(I_j^-).$$

Proof. The scalar fiber equations are exactly the same as in Section 6; only the ambient measure changes from Lebesgue measure to μ . By Corollary 4.6, the task fiber kernel is one-dimensional exactly on I_j^+ and the adjoint task fiber kernel is one-dimensional exactly on I_j^- . The measurable rank-one fields constructed in Lemma 6.7 are pointwise objects and therefore remain measurable with respect to μ . Their traces are the μ -measures of the support intervals. The nuisance channels are strictly positive and contribute no kernel. $\square$

Theorem 8.4. *Assume that the path $u \mapsto A_u^{\mu}$ satisfies the hypotheses of the Azzali–Wahl APS theorem in the semifinite algebra $\mathcal{N}_{\mu}$. Then*

$$\text{sf}_{\text{Tr}_{\mu}}\{A_u^{\mu}\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}}_{\mu}}(D_{\mu, \text{APS}}) = \int_{\mathbb{R}} \nu_T^{\mu}(\lambda) d\mu(\lambda) = \sum_{j=1}^{r_T} \mu(I_j^+) - \sum_{j=1}^{r_T} \mu(I_j^-).$$

Moreover the nuisance channels contribute trivially.

Proof. By Proposition 8.3,

$$\text{Ind}_{\widetilde{\text{Tr}}_{\mu}}(D_{\mu, \text{APS}}) = \sum_{j=1}^{r_T} \mu(I_j^+) - \sum_{j=1}^{r_T} \mu(I_j^-).$$

Since the path satisfies the Azzali–Wahl hypotheses by assumption,

$$\text{sf}_{\text{Tr}_{\mu}}\{A_u^{\mu}\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}}_{\mu}}(D_{\mu, \text{APS}}).$$

Finally,

$$\nu_T^{\mu}(\lambda) = \sum_{j=1}^{r_T} 1_{I_j^+}(\lambda) - \sum_{j=1}^{r_T} 1_{I_j^-}(\lambda)$$

for μ -almost every λ , so integration gives the stated formula. $\square$

Corollary 8.5. *If every task channel α_j is monotone, then*

$$\text{sf}_{\text{Tr}_{\mu}}\{A_u^{\mu}\}_{u \in [0,1]} = \sum_{\alpha_j \text{ increasing}} \mu((- \alpha_j(1), - \alpha_j(0))) - \sum_{\alpha_j \text{ decreasing}} \mu((- \alpha_j(0), - \alpha_j(1))).$$

Proof. If α_j is increasing, then $I_j^- = \emptyset$ and

$$I_j^+ = (-\alpha_j(1), -\alpha_j(0)).$$

If α_j is decreasing, then $I_j^+ = \emptyset$ and

$$I_j^- = (-\alpha_j(0), -\alpha_j(1)).$$

Substitute into Theorem 8.4. □

8.2 A radial hyperbolic model

Let

$$X = \mathbb{H}^d = SO_0(d, 1)/SO(d), \quad \rho := \frac{d-1}{2}.$$

By External input (E), there is a unitary spherical transform on the radial subspace

$$\mathcal{F}_{sph} : L_{\text{rad}}^2(X) \rightarrow L^2([0, \infty), d\mu_\rho).$$

For vector-valued radial functions, we use its tensorization

$$\mathcal{F}_{sph}^{(m)} := \mathcal{F}_{sph} \otimes 1_{\mathbb{C}^m} : L_{\text{rad}}^2(X, \mathbb{C}^m) \rightarrow L^2([0, \infty), \mathbb{C}^m; d\mu_\rho).$$

Define

$$s_\rho(\lambda) := \sqrt{\lambda^2 + \rho^2}.$$

Set

$$\mathcal{H}_\rho := L^2([0, \infty), \mathbb{C}^m; d\mu_\rho), \quad \mathcal{N}_\rho := L^\infty([0, \infty), \mu_\rho; M_m(\mathbb{C})),$$

with

$$\text{Tr}_\rho(M_{q(\lambda)}) = \int_0^\infty \text{tr}(q(\lambda)) d\mu_\rho(\lambda), \quad \widetilde{\text{Tr}}_\rho := \text{Tr}_{B(L^2([0,1]))} \bar{\otimes} \text{Tr}_\rho.$$

Let

$$\tau_j \in C^1([0, 1], \mathbb{R}), \quad 1 \leq j \leq r_T,$$

and assume each τ_j is constant near the endpoints. Let

$$\eta_k \in C([0, 1], [\eta_0, \infty)), \quad \eta_0 > 0, \quad 1 \leq k \leq r_N,$$

and assume each η_k is constant near the endpoints.

Define, on the radial subspace of $L^2(X, \mathbb{C}^m)$,

$$A_u^\rho = \bigoplus_{j=1}^{r_T} (\tau_j(u)I - (-\Delta_X)^{1/2}) \oplus \bigoplus_{k=1}^{r_N} ((-\Delta_X)^{1/2} + \eta_k(u)I).$$

Under $\mathcal{F}_{sph}^{(m)}$, this is unitarily equivalent to multiplication by

$$a_u^\rho(\lambda) = \text{diag}\left(\tau_1(u) - s_\rho(\lambda), \dots, \tau_{r_T}(u) - s_\rho(\lambda), s_\rho(\lambda) + \eta_1(u), \dots, s_\rho(\lambda) + \eta_{r_N}(u)\right).$$

Define the exceptional set

$$E_\rho := \{\lambda \geq 0 : s_\rho(\lambda) = \tau_j(0) \text{ or } s_\rho(\lambda) = \tau_j(1) \text{ for some } j\}.$$

Since s_ρ is strictly increasing and the endpoint set is finite, E_ρ is finite, and therefore $\mu_\rho(E_\rho) = 0$.

For each task channel j , define

$$J_j^+ := \{\lambda \geq 0 : \tau_j(0) - s_\rho(\lambda) < 0 < \tau_j(1) - s_\rho(\lambda)\},$$

$$J_j^- := \{\lambda \geq 0 : \tau_j(0) - s_\rho(\lambda) > 0 > \tau_j(1) - s_\rho(\lambda)\}.$$

Define the task density

$$\nu_T^\rho(\lambda) = \sum_{j=1}^{r_T} \frac{1}{2} \left(\operatorname{sgn}(\tau_j(1) - s_\rho(\lambda)) - \operatorname{sgn}(\tau_j(0) - s_\rho(\lambda)) \right),$$

for $\lambda \notin E_\rho$, and arbitrarily on the null set E_ρ .

Lemma 8.6. *Let $\mathcal{D}_\rho$ denote the domain of the radial operator $(-\Delta_X)^{1/2}$. Then each A_u^ρ is selfadjoint on the common domain $\mathcal{D}_\rho$. Moreover, for all $u, v \in [0, 1]$, the difference $A_u^\rho - A_v^\rho$ is bounded.*

Proof. Each task channel

$$\tau_j(u)I - (-\Delta_X)^{1/2}$$

is a bounded selfadjoint perturbation of $-(-\Delta_X)^{1/2}$, while each nuisance channel

$$(-\Delta_X)^{1/2} + \eta_k(u)I$$

is a bounded selfadjoint perturbation of $(-\Delta_X)^{1/2}$. Hence all channels have the common domain $\mathcal{D}_\rho$. The difference $A_u^\rho - A_v^\rho$ is channelwise given by scalar bounded differences $\tau_j(u) - \tau_j(v)$ and $\eta_k(u) - \eta_k(v)$. $\square$

Lemma 8.7. *For every $u \in [0, 1]$,*

$$(A_u^\rho \pm i)^{-1} \in K(\mathcal{N}_\rho).$$

Proof. Under the spherical transform, the path is a multiplier path with symbol $a_u^\rho(\lambda)$. For the task channels,

$$|(\tau_j(u) - s_\rho(\lambda) \pm i)^{-1}| = \frac{1}{\sqrt{(\tau_j(u) - s_\rho(\lambda))^2 + 1}}.$$

If this is at least ε , then

$$|\tau_j(u) - s_\rho(\lambda)| \leq \sqrt{\varepsilon^{-2} - 1}.$$

Since $s_\rho(\lambda) \rightarrow \infty$ as $\lambda \rightarrow \infty$, this confines λ to a bounded interval. For the nuisance channels,

$$|(s_\rho(\lambda) + \eta_k(u) \pm i)^{-1}| = \frac{1}{\sqrt{(s_\rho(\lambda) + \eta_k(u))^2 + 1}},$$

and the same conclusion is immediate. Since bounded intervals have finite μ_ρ -measure by External input (E), the compactness criterion from Section 8 applies. $\square$

Lemma 8.8. *The function ν_T^ρ belongs to $L^1([0, \infty), d\mu_\rho)$.*

Proof. For each fixed j , the function

$$\lambda \mapsto \frac{1}{2} \left(\operatorname{sgn}(\tau_j(1) - s_\rho(\lambda)) - \operatorname{sgn}(\tau_j(0) - s_\rho(\lambda)) \right)$$

is the indicator function of the bounded interval J_j^+ , or minus the indicator of J_j^- , or zero. Since bounded intervals have finite μ_ρ -measure, each such function lies in $L^1(d\mu_\rho)$. Summing over finitely many channels preserves integrability. $\square$

For $\lambda \notin E_\rho$, define the fiberwise APS operator

$$D_{\lambda, \text{APS}}^\rho := \frac{d}{du} + a_u^\rho(\lambda)$$

on $L^2([0, 1], \mathbb{C}^m)$ with APS boundary conditions determined by the endpoint matrices. Define the global APS operator

$$D_{\rho, \text{APS}} := \int_{[0, \infty)}^\oplus D_{\lambda, \text{APS}}^\rho d\mu_\rho(\lambda),$$

with domain

$$\text{Dom}(D_{\rho, \text{APS}}) := \left\{ f \in \int_{[0, \infty)}^\oplus \text{Dom}(D_{\lambda, \text{APS}}^\rho) d\mu_\rho(\lambda) : \int_0^\infty \|D_{\lambda, \text{APS}}^\rho f(\lambda)\|_{L^2([0, 1], \mathbb{C}^m)}^2 d\mu_\rho(\lambda) < \infty \right\}.$$

Lemma 8.9. *Fix a task channel j . For $\lambda \in J_j^+$, define*

$$\psi_{j, \lambda}^\rho(u) := \exp\left(-\int_0^u (\tau_j(s) - s_\rho(\lambda)) ds\right) \in L^2([0, 1]).$$

Then the map

$$\lambda \mapsto \psi_{j, \lambda}^\rho$$

is continuous from J_j^+ into $L^2([0, 1])$. In particular, the normalized vectors give a measurable rank-one projection field $p_j^{\rho, +}(\lambda)$ on J_j^+ . Similarly, on J_j^- one obtains a measurable rank-one projection field $p_j^{\rho, -}(\lambda)$ for the adjoint equation.

Proof. For fixed $u \in [0, 1]$,

$$\lambda \mapsto \psi_{j, \lambda}^\rho(u) = \exp\left(-\int_0^u \tau_j(s) ds\right) e^{us_\rho(\lambda)}$$

is continuous because s_ρ is continuous. On every compact subinterval of J_j^+ , the family is uniformly bounded by an $L^2([0, 1])$ -function independent of λ , so dominated convergence implies continuity in $L^2([0, 1])$. Normalization preserves measurability. The adjoint case is identical. $\square$

Proposition 8.10. *For the hyperbolic APS operator one has, channel by channel,*

$$\widetilde{\text{Tr}}_\rho(P_{\ker D_{\rho, \text{APS}, j}}) = \mu_\rho(J_j^+), \quad \widetilde{\text{Tr}}_\rho(P_{\ker D_{\rho, \text{APS}, j}^*}) = \mu_\rho(J_j^-).$$

The nuisance channels contribute trivially to both kernels. Consequently, whenever $D_{\rho, \text{APS}}$ is Breuer–Fredholm,

$$\text{Ind}_{\widetilde{\text{Tr}}_\rho}(D_{\rho, \text{APS}}) = \sum_{j=1}^{r_T} \mu_\rho(J_j^+) - \sum_{j=1}^{r_T} \mu_\rho(J_j^-).$$

Proof. Under the spherical transform, the radial hyperbolic model becomes a direct-integral multiplier model with scalar task coefficients

$$u \mapsto \tau_j(u) - s_\rho(\lambda),$$

and nuisance coefficients

$$u \mapsto s_\rho(\lambda) + \eta_k(u).$$

By Corollary 4.6, the task fiber kernel is one-dimensional exactly on J_j^+ , and the adjoint task fiber kernel is one-dimensional exactly on J_j^- . By Lemma 8.9, the corresponding projection fields are measurable. Their traces are the μ_ρ -measures of the support intervals. For the nuisance channels, the coefficients are strictly positive for all (u, λ) , so the fiber kernels vanish. $\square$

Theorem 8.11. *Assume that the path $(A_u^\rho)_{u \in [0,1]}$ satisfies the hypotheses of the Azzali–Wahl APS theorem in the semifinite algebra $\mathcal{N}_\rho$. Then*

$$\text{sf}_{\text{Tr}_\rho} \{A_u^\rho\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}_\rho}}(D_{\rho, \text{APS}}) = \int_0^\infty \nu_T^\rho(\lambda) d\mu_\rho(\lambda) = \sum_{j=1}^{r_T} \mu_\rho(J_j^+) - \sum_{j=1}^{r_T} \mu_\rho(J_j^-).$$

In particular, the nuisance channels contribute trivially.

Proof. Under the spherical transform, the radial hyperbolic model is unitarily equivalent to the multiplier path

$$u \mapsto M_{a_u^\rho(\lambda)}$$

on

$$L^2([0, \infty), \mathbb{C}^m; d\mu_\rho).$$

By Lemma 8.6, this path has a common domain and is continuous in graph norm. By Lemma 8.7,

$$(A_u^\rho \pm i)^{-1} \in K(\mathcal{N}_\rho) \quad (\forall u \in [0, 1]).$$

Since the path is constant near the endpoints, the APS theorem of Azzali–Wahl applies and yields

$$\text{sf}_{\text{Tr}_\rho} \{A_u^\rho\}_{u \in [0,1]} = \text{Ind}_{\widetilde{\text{Tr}_\rho}}(D_{\rho, \text{APS}}).$$

Now Proposition 8.10 gives

$$\text{Ind}_{\widetilde{\text{Tr}_\rho}}(D_{\rho, \text{APS}}) = \sum_{j=1}^{r_T} \mu_\rho(J_j^+) - \sum_{j=1}^{r_T} \mu_\rho(J_j^-).$$

Finally, by definition of ν_T^ρ ,

$$\nu_T^\rho(\lambda) = \sum_{j=1}^{r_T} 1_{J_j^+}(\lambda) - \sum_{j=1}^{r_T} 1_{J_j^-}(\lambda)$$

for μ_ρ -almost every λ . By Lemma 8.8, this function is integrable, and therefore

$$\int_0^\infty \nu_T^\rho(\lambda) d\mu_\rho(\lambda) = \sum_{j=1}^{r_T} \mu_\rho(J_j^+) - \sum_{j=1}^{r_T} \mu_\rho(J_j^-).$$

Combining the three displayed equalities proves the theorem. $\square$

Corollary 8.12. *If a task channel τ_j is monotone increasing, then*

$$J_j^- = \emptyset$$

and

$$J_j^+ = \{\lambda \geq 0 : \tau_j(0) < s_\rho(\lambda) < \tau_j(1)\}.$$

If a task channel τ_j is monotone decreasing, then

$$J_j^+ = \emptyset$$

and

$$J_j^- = \{\lambda \geq 0 : \tau_j(1) < s_\rho(\lambda) < \tau_j(0)\}.$$

Hence

$$\text{sf}_{\text{Tr}_\rho} \{A_u^\rho\} = \sum_{\tau_j \text{ increasing}} \mu_\rho(J_j^+) - \sum_{\tau_j \text{ decreasing}} \mu_\rho(J_j^-).$$

Proof. This is immediate from the definition of $J_j^\pm$ and Theorem 8.11. $\square$

9 Conclusion

The purpose of this paper has been deliberately limited. Rather than presenting a full proper-action Jacobian-chain index theory in one step, we have isolated a rigorous core consisting of compact and noncompact structured models in which selective spectral flow can be computed exactly.

At the scalar level, Theorem 4.4 shows that the relevant APS index is governed by a signed endpoint formula. This elementary result propagates upward into all later constructions. On the compact side, Theorem 5.6 yields an explicit S^1 -representation-valued propagation formula, while Theorem 7.4 upgrades this picture to a genuinely nonabelian $SO(3)$ -equivariant model on S^2 . On the continuous-spectrum side, Theorem 6.9 shows that the total semifinite spectral flow is the integral of an explicit task spectral-flow density, and Theorem 8.11 places this phenomenon in a genuinely geometric noncompact setting through a radial model on real hyperbolic space.

Across all these models, a common pattern emerges: task channels are allowed to contribute protected propagation through explicit spectral crossings, whereas nuisance channels satisfying nonresonance or uniform-gap conditions contribute nothing. In this sense, the paper provides rigorous no-contribution theorems for nuisance channels and explicit propagation formulas for task channels.

We stress again that the present work does not yet amount to a full general theory. In particular, it does not establish a complete proper-action Jacobian-chain index theory, nor does it apply directly to arbitrary deep neural networks. What it does provide is a mathematically controlled core from which a broader program can reasonably proceed.

There are at least three natural directions for continuation. First, one should connect the compact and radial models proved here to a genuine proper-action framework, with values in

$$KK(C_0(G/K) \rtimes G, \mathbb{C}),$$

as suggested by the Hochs–Song picture [HS17, HS18]. Second, one should refine the scalar semifinite spectral-flow formulas to $K_0(J)$ -valued classes in the spirit of Kaa–Nest–Rennie [KNR12]. Third, one should develop computable diagnostics and proxies for the representation-resolved and mode-resolved propagation quantities introduced here, in order to make contact with concrete structured models in geometric deep learning.

From the perspective of deep learning, the main conceptual contribution of the paper is not a ready-made algorithmic recipe, but a shift in what one tries to measure. Instead of asking only whether a model is globally stable, one is led to ask: which symmetry sectors carry protected propagation, which continuous modes contribute spectral flow, and which nuisance channels are forced into a no-crossing regime? In structured geometric settings, these questions admit exact answers. We regard that as a mathematically meaningful step toward a broader Jacobian-chain index theory for deep learning.

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